Fachhochschule Dortmund
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Information Processing and Data Analytics Report

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European Management in

Project Management

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1. MANAGEMENT SUMMARY

The importance of using big data to drive organizational decision-making in today's business environment cannot be exaggerated. Through this report, we analyse SPAR's 2009 sales data and provide recommendations for business improvement initiatives. Some recommendations discussed in this report include optimizing sales for future growth, conducting strategic analyses for optimal results, analysing Waidhofen success and implementing it in another region, the recommendation to conducting regional seasonal analysis, enhancing inventory control, and addressing performance data anomalies. Our analysis led to interesting findings on the low sales performance of cosmetics products, and for this, we recommended some cost-effective marketing strategies. Understanding that customer feedback and needs are important for business success, we also investigated general recommendations to meet customer needs. Some of these recommendations include seeking continuous improvement from customer feedback, digital engagement and the delivery of personalized recommendations aimed at boosting SPAR's visibility and customer appeal, considering digital retailing and home services, and implementing customer loyalty cards. The overall recommendations made in this report are aimed not only at enhancing SPAR's profitability but also at attracting and retaining customers by ensuring a smooth and enjoyable shopping experience.

2. INTRODUCTION

This document relies on a detailed examination of SPAR's information in Lower Austria as a case study, with the goal of providing recommendations for enhancing business performance. The dataset for the analysis comprises of 2009 turnover data per product group per week for SPAR supermarkets in Lower Austria. The approach involved consolidating weekly data into a single table, utilizing MS-Excel for ad-hoc comparisons, focusing on three product groups out of 39. In addition to the SPAR data, reliable information from sources like Google, Spar website and public reports was integrated for a comprehensive analysis, leading to recommendations in this management report.

2.1. Background

Spar Österreichische Warenhandels-AG (brand name: SPAR Austria Group), headquartered in Salzburg's Taxham district, is the leading Austrian trading company with a strong presence in Austria, Italy, Slovenia, Hungary, and Croatia. Since 2020, it has dominated the Austrian food retail market and is a key player in the international Spar retail chain (Spar Österreichische Warenhandels-AG).

2.2. Key figures

The Spar Austria Group, boasting approximately 1,580 locations (SPAR, EUROSPAR, INTERSPAR and Maximarkt) across Austria as of December 31, 2021, features a significant portion operated by independent retailers who source products from Spar wholesalers. Globally, the group managed around 3,300 establishments in 2021, including 250 in sports retail and 30 shopping centres, with a workforce of about 90,000 employees in Austria and abroad. As Austria's largest private employer with nearly 50,000 employees and a leading private apprenticeship trainer, Spar holds a dominant 36.00% market share in the Austrian food retail sector in 2021, surpassing Rewe International AG (Spar Österreichische Warenhandels-AG).

2.3. Segment and Store Types:

Spar Austria organizes its supermarkets into various kinds of stores. These stores come in different sizes and offer various products. This helps Spar provide a range of options that suit the needs of different customers.

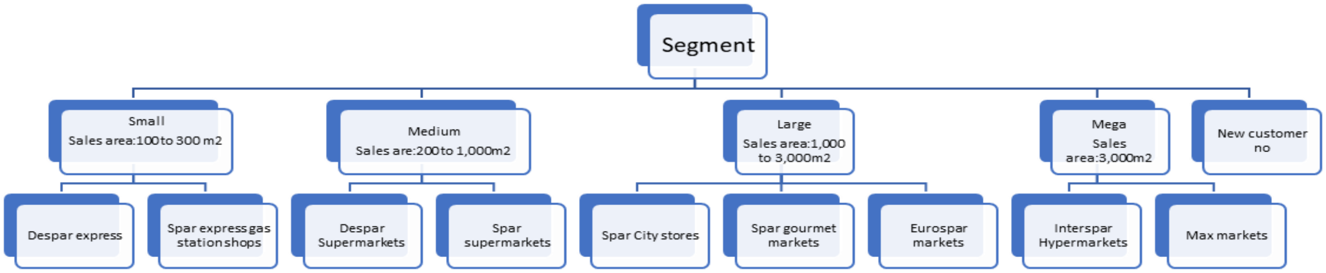

Figure 1 - Segment and Store Types. Source: spar-international.com

3. METHODOLOGY

In this section, we discuss how the data provided (SPAR Lower Austria Basic Data) and other additional sources were used to gain meaningful insights. We delve into the approach taken towards data cleaning, processing and analysis.

3.1. Introduction

The data provided was in Microsoft excel format and included the turnover with and without value-added tax (VAT), from every product group in every SPAR store location in lower Austria for the year 2009. The data provided consisted of 3 categories. The first category consisted of 53 Microsoft Excel sheets representing the turnover of 53 weeks for 39 product groups in specific supermarket branches. These sheets include seven columns: week, segment, region, customer number, product group, turnover without VAT and turnover with VAT. The “Week” column in the data represents the number of weeks of the data, while the “Segment” column shows the varied sizes of the SPAR supermarket branches. The segment is divided into five categories: Mega, Large, Middle, Small and New Customer Number. The new customer is the newly opened customer in the referenced year. The “Region” column also categorises complete SPAR supermarket areas in Lower Austria, encompassing 21 city and rural areas. The “product group” column is a collection of products sold by SPAR supermarket. It consists of a total product group and 39 different product types. Lastly, are the turnover columns. This column is the turnover value, and it consists of 2 columns: turnover with VAT and turnover without VAT.

The second sheet category is the total turnover in every SPAR region and store in 2009. This sheet consists of 8 columns, including reference year, segment, region, customer number, “total” product group, turnover without VAT, turnover with VAT, and percentage of turnover from last year. Last, the third sheet category contains explanations regarding the Microsoft Excel data and the list of product groups.

3.2. Merge Weekly Turnover

Our group decided to merge all 53-week turnover sheets into one sheet using a macro in Microsoft Excel to make it easier to know the trend of the SPAR supermarket. The script used in this course was adapted from ExcelTip.com. Find below the script:

```
Sub Merge_Sheetsv2()
    Dim startRow, startCol, lastRow, lastCol As Long
    Dim headers As Range
    Dim wb As Workbook
    Dim ws, combinedSheet As Worksheet
    Dim wsCount As Integer

    'create a new workbook
    Set wb = ThisWorkbook
    Set combinedSheet = wb.Sheets.Add(After:=wb.Sheets(wb.Sheets.Count))
    combinedSheet.Name = "CombinedData"

    'loop through all sheets
    For Each ws In wb.Sheets
        'stop if ws is equal CobinedData
        If ws.Name <> "CombinedData" Then
            'find the last row
            lastRow = combinedSheet.Cells(combinedSheet.Rows.Count, 1).End(xlUp).Row
            'get data from each worksheet and copy it into Master sheet
            ws.UsedRange.Copy combinedSheet.Cells(lastRow + 1, 1)
        End If
    Next ws

    Worksheets("CombinedData").Activate
End Sub
```

Figure 2 - Source Code for Merging Sheets. Adapted from ExcelTip.com

3.3. Cleaning Data and Add Additional Data

After merging 53 weekly turnover sheets into one combined sheet, we changed the blank value into “0” to prevent mixed data type in our data and make it easier to analyse the data and use the pivot function. Furthermore, we added three additional columns (month, population, and notes). The “month” column is the number of months based on a given week. This column is used for the revenue per segment and growth rate per product category. To get the number of months, we used the formula:

=DATE (2009 ; 1 ; 1) + ([week cell reference] – 1) * 7

The “population” column is the population in specific regions in lower Austria. The data that we used came from worldpopulationreview.com and combined with Wikipedia population data for 12 cities (Bruck, Gänserndorf, Gmünd, Krems, Lilienfeld, Melk, Mödling, Scheibbs, St. Pölten, Waidhofen/T, Wien Umgebung, Zwettl) that were not available on the website. This column is used as additional information to describe trends in some regions. For example, Wien Umgebung has a high turnover without any specific reason. With the population column, we can assume that the trend happens because of the population in that region. The second additional column is notes, filled with our group comments about unrealistic numbers or data anomalies.

Our group also has a sheet for major events celebrated in Austria. This data is used for seasonal analysis that we identify patterns, influencing factors and opportunities to improve sales performance.

3.4. Analysing Data

For the analysis three pivot sheets were created as seen below:

- the weekly turnover without VAT by segment filtered by product group.
- the weekly turnover without VAT by regions filtered by product group as well as segment.
- the weekly turnover without VAT by customer number filtered by product group and region.

Afterwards, the pivot data was visualised in a line graph chart to understand the trend and draw a conclusion from the graph. The process of analysis proceeded manually based on the created graph.

Furthermore, a pivot of monthly turnover without VAT by segment and filtered by product group was created to know the revenue of selected product groups, wine, sweets, and cosmetics, per segment. This data was visualised into a bar chart because it can provide transparent displays of the revenue of selected products per month. Based on this pivot data, the growth rate data of selected products could be produced. The formula used to get the growth rate is shown below:

*= (([Current Month Revenue in Segment One] - [Previous Month Revenue in Segment One]) / [Previous Month Revenue in Segment One]) * 100.*

Lastly, the percentage of total sales in wine, sweets, and cosmetics was created to determine how much turnover was generated by segments. This data is compared with the number of stores in each segment, and both data are displayed in a pie chart.

4. ANALYSIS RESULTS

In this section, we discuss in detail the analysis carried out on the data provided. Our analysis is divided into three parts: seasonal analysis, geographical analysis, and revenue trend analysis.

4.1. Seasonal analysis

An analysis of the data was made considering the time of year to identify peak and trough periods. For this purpose, three product types were selected: sweets, wine, and cosmetics. This analysis did not only provide a retrospective view of sales trends over the last year, but will also identify patterns, influencing factors and opportunities to improve sales performance.

First, wine sales throughout the year were analysed, classifying these sales according to the different types of existing supermarkets: large, mega, middle, small and new supermarkets. The figure below shows the turnover vs the weeks with the wine data.

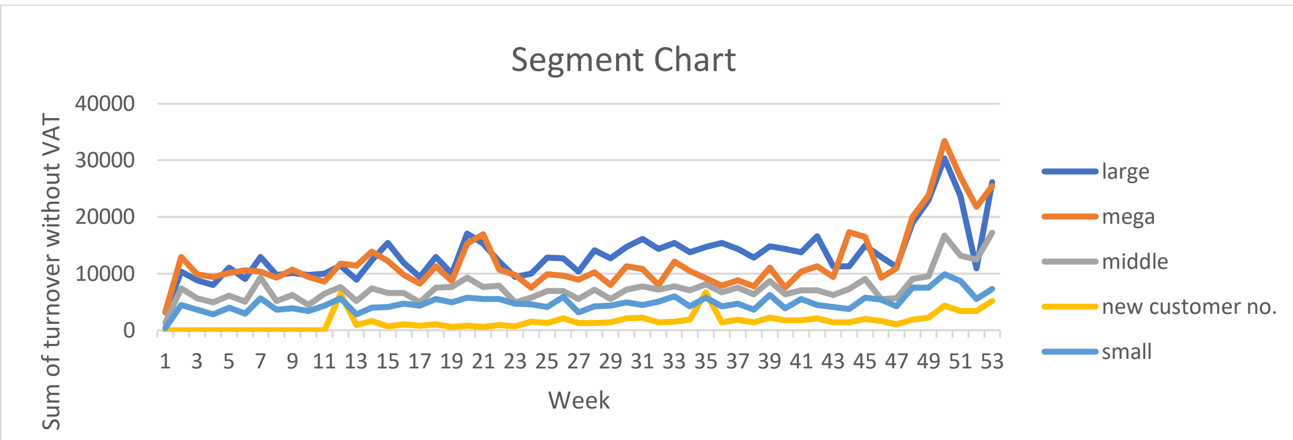

Figure 3 - Wine sales

In this graph all segments share a similar pattern in terms of wine sales during the year, except for weeks 15 and 21 where a peak can be seen in the large and mega segments. Week 15 (April 6th to 12th) corresponds to Easter week, which would justify the increase in wine sales. 40 days after Easter Sunday, which in 2009 was 12 April, the day of the Ascension is celebrated, this day in 2009 was 21 May, which coincides with the 21st week of the year. The Assumption Day is a public holiday in several countries, among them Austria.

The highest points of the graph occur with the mega and large segments in the 50th week. This week corresponds to the week from 7 to 13 December. The increase in wine sales at this time of the year may be due to the impact of Christmas, as it is common to visit Christmas markets and drink cups of wine there.

The second product that was analysed were the sweets. This data is shown in Figure 4:

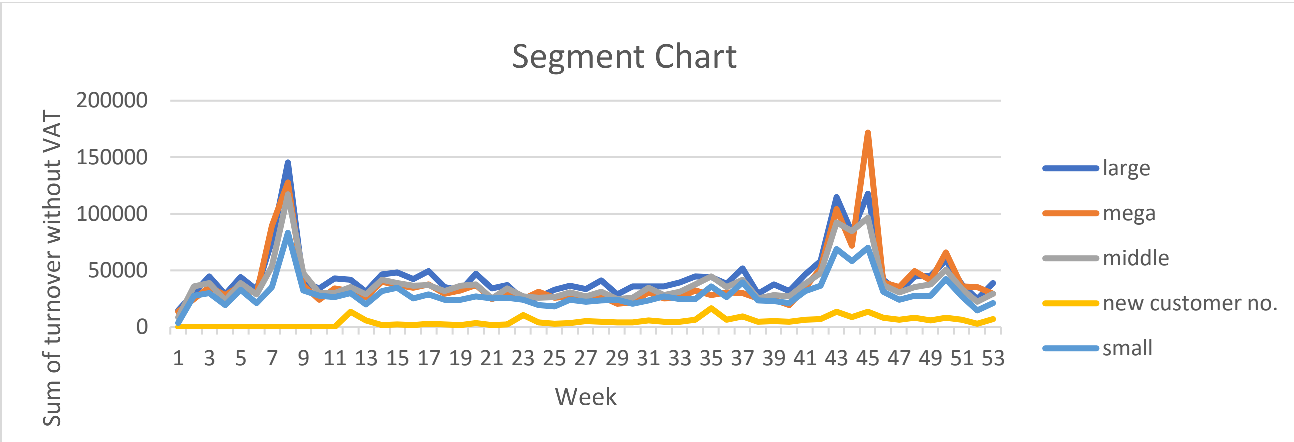

Figure 4 - Sweets sales

In general, the sales of the products show very similar patterns when comparing the different segments. Therefore, the largest peaks found during the year were analysed. The first peak occurs around week 8, which corresponds to the week from 16 to 22 February. This increase in sales of sweets may be due to two different events on these dates. On 14 February, Valentine's Day is celebrated, and it is a tradition to give gifts and sweets to couples. The sale of sweets may also be affected by the carnival season. The Saturday before Ash Wednesday (February 2), the Carnival parades are held. Children will dress up in costumes and teenagers or adults go to fancy dress parties. In small communities, there are parades and processions, and it is common to give sweets. The second peak occurs between the weeks 43 and 45, corresponding to the ending October and the beginning of November. Halloween is celebrated on 31 October. On this date it is common for children to dress up in costumes and go trick-or-treating. On the other hand, the 11th of November is St. Martin's Day. The nights before and on the night of 11 November, children walk in processions carrying lanterns, which they made in school.

The last product to be analysed was cosmetics, and this information is shown in Figure 5.

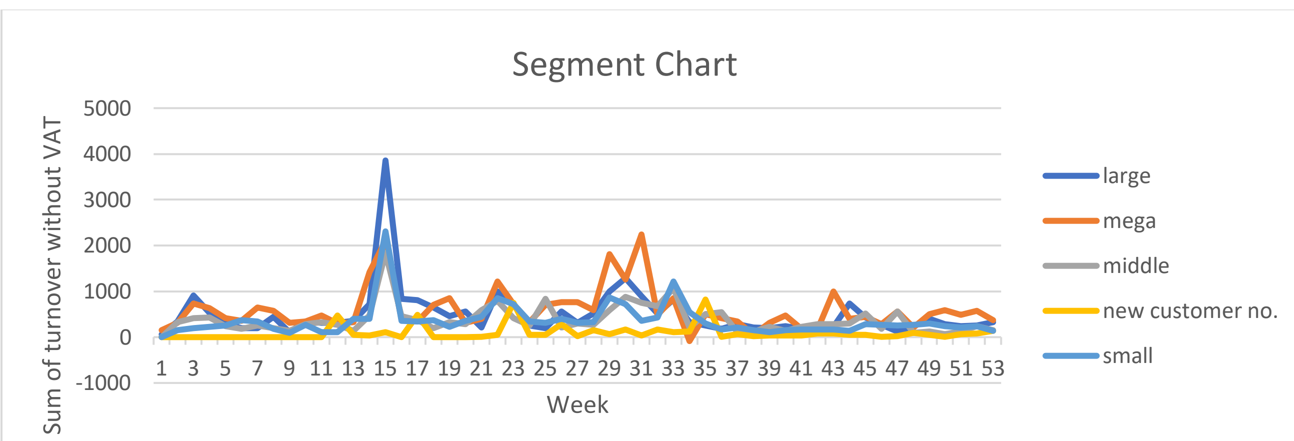

Figure 5 - Cosmetics sales

Two peaks of interest appear in this graph. The first of these is a peak that coincides with all segments but is highest in large supermarkets. This occurs in week 15, corresponding to the days from 6 to 12 April, Easter week in 2009. The rise in sales of cosmetic products may be related to this event because there are many different customs with regional significance held in Austria in the week that follows, and passion plays are organised in some communities. Also, Easter eggs are painted and decorated, and that could justify using some products for the decoration.

The second peak detected it is a little bit confusing. This occurs mostly with mega supermarkets and the peak rises in week 29, falls in week 30 and rises again in week 31. It is not clear to us what could have happened on these dates, as they coincide with the month of July and there is no significant event to highlight. Even so, it could be because at that time there was an event or festival in one of the areas where some mega supermarkets are located.

4.2. Geographic Analysis

The geographic analysis aims to analyse the turnover data by region to identify geographic areas with strong, weak or unique performance.

4.2.1. Geographic Analysis - Total

The given data consisted of 39 product groups in every lower Austria region in 2009. This is captured in the figure below:

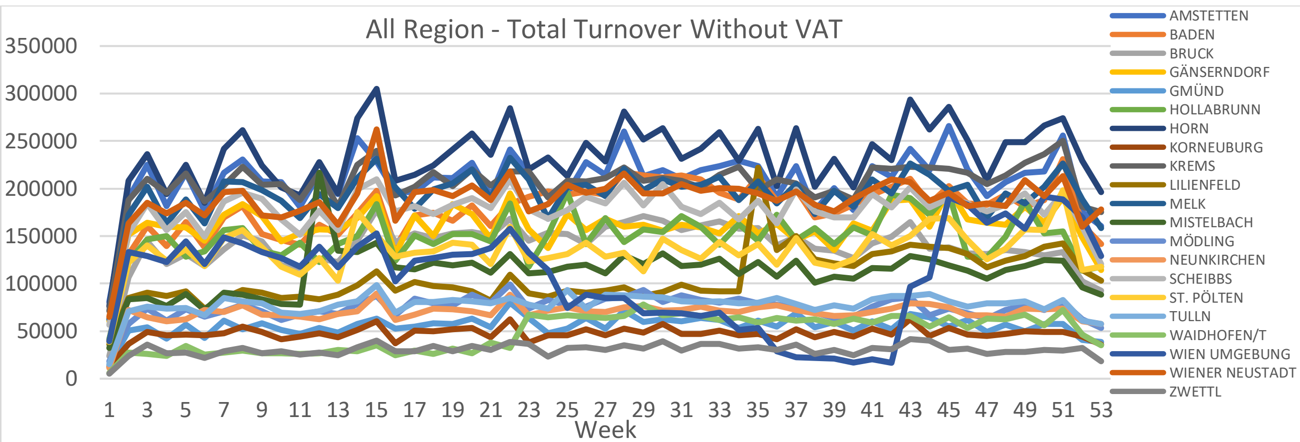

Figure 6 - All Region - Total Turnover Without VAT

The data of all regions with the total turnover without VAT (Figure 6) shows a similar trend from week to week in almost all regions. However, an exciting region had a unique performance during 2009, Waidhofen/t, with a sudden increase in week 23.

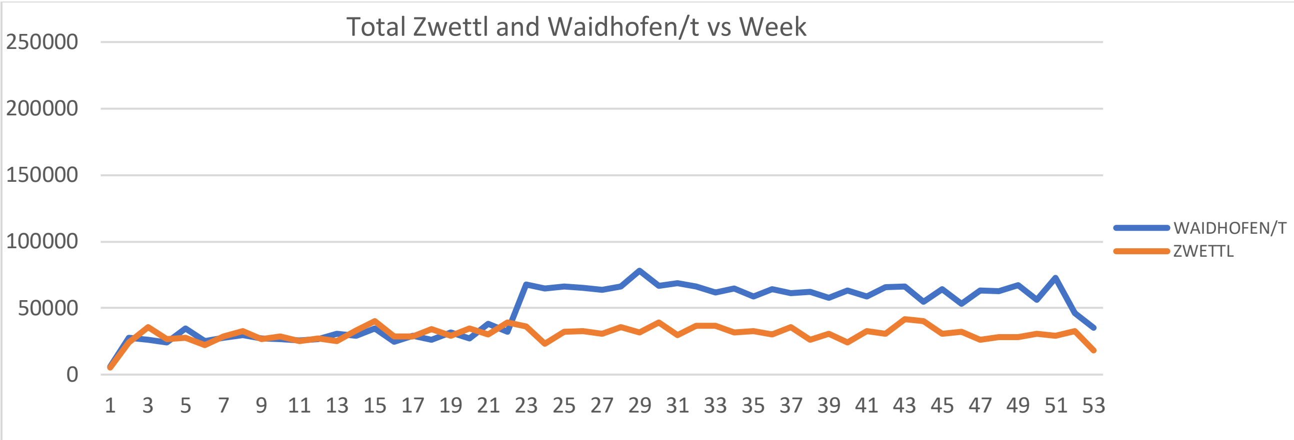

Figure 7 - Total Turnover Without VAT in Zwettl and Waidhofen/t vs Week

Looking at the data of total turnover without VAT in Waidhofen/t and Zwettl (Figure 7), both cities follow almost a similar trend until Week 22. A week later, Waidhofen/t experienced a considerable increase that continued steadily until Week 53. As a result, Waidhofen/t took a different path from Zwettl starting in Week 23, showing consistent growth and stability.

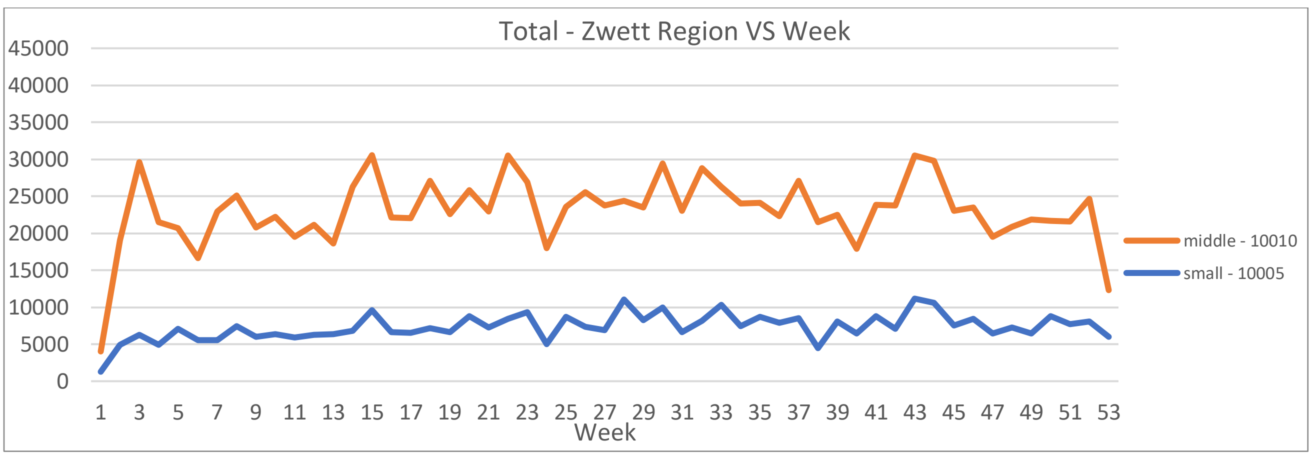

Figure 8 - Total - Zwett Region VS Week

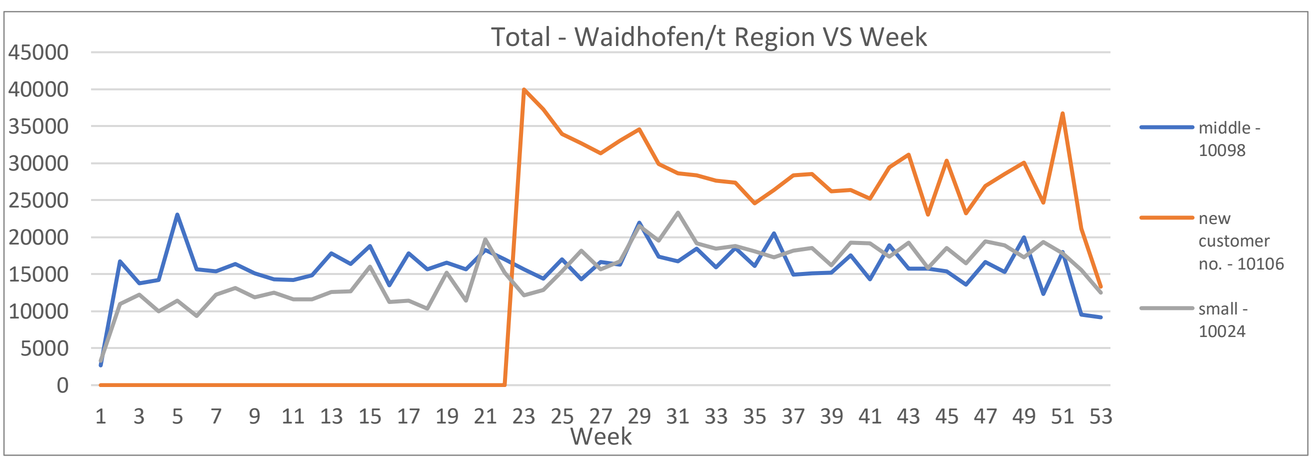

Figure 9 - Total - Waidhofen/t Region VS Week

A deeper investigation into the data shows that Waidhofen/t (Figure 9) experienced a significant increase in total turnover without VAT starting from Week 23 because of the opening of a new store in that region. This new store boosts the total turnover and significantly increases Waidhofen/t's performance. On the other hand, Zwettl (Figure 4) did not show a similar trend in the total turnover without VAT because there was no event in that region during this period, such as opening a new store. It explains why Zwettl's wine sales remained consistent without any sudden increases.

4.2.2. Geographic Analysis - Wine

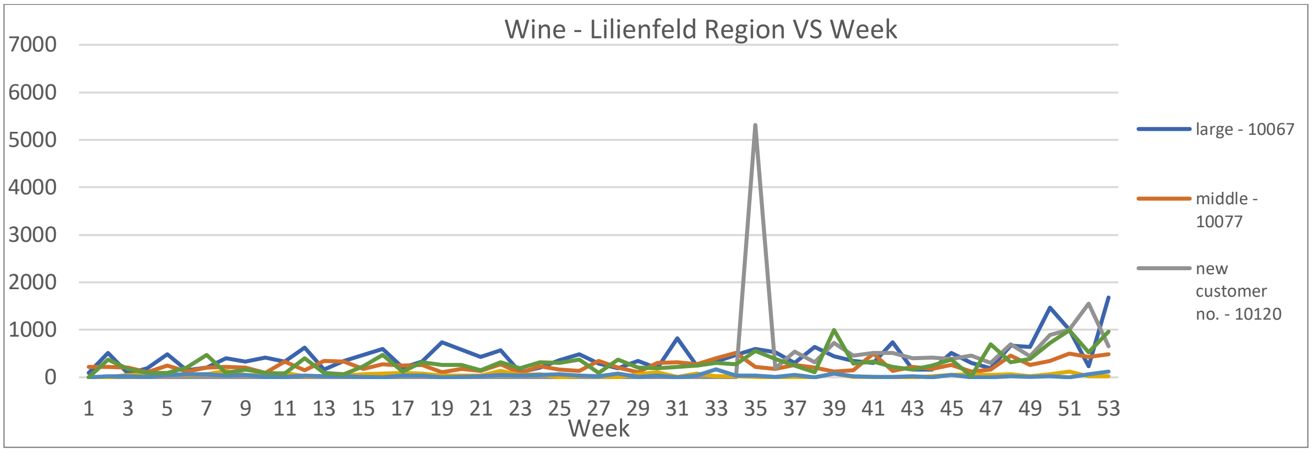

Figure 10 Wine - Lilienfeld Region VS Week

When looking at the wine turnover without VAT data, a fascinating trend appeared in Lilienfeld. In Week 35, there was a remarkable and sudden rise in turnover without VAT, which also directly correlated with opening a new store in that location. This new store's launch was a catalyst, causing a significant spike in wine turnover, specifically within Lilienfeld.

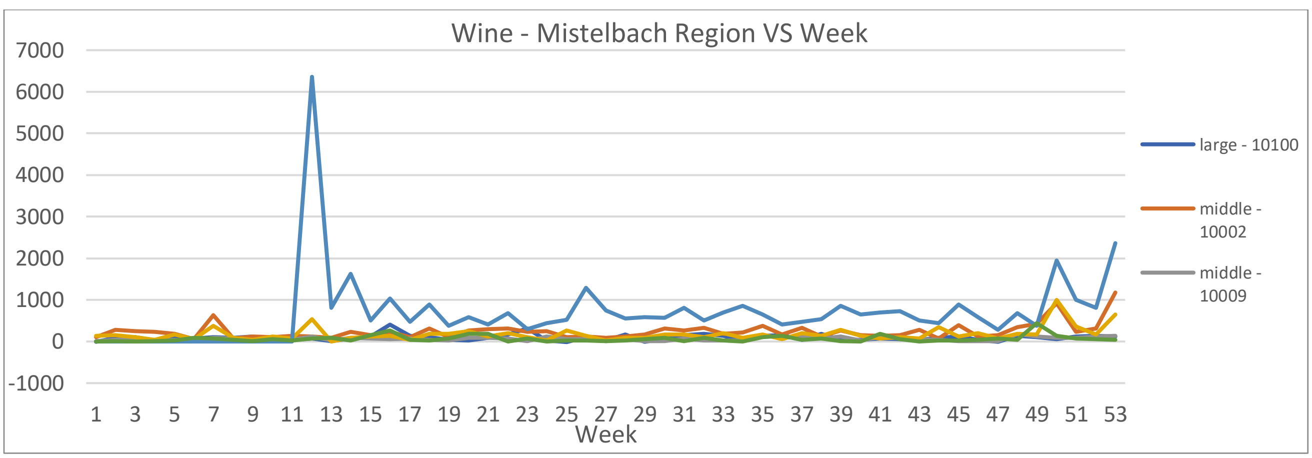

Figure 11 Wine - Mistelbach Region VS Week

Like Lilienfeld, Mistelbach shows a comparable trend in wine sales because of the opening of a new store. In Week 12, a sudden and significant increase in wine turnover was directly linked to the store's launch. Like Lilienfeld, this trend was short-lived, lasting only one week, and sales remained relatively steady until Week 49. Towards the end of the year, sales slightly increased, aligning with the trend in other stores. There is also a negative turnover in week 25 in a large segment in Mistelbach with -13,7. We have chosen not to remove this data point immediately. Instead we choose to confirm its accuracy before making any adjustments.

4.2.3. Geographic Analysis - Sweets

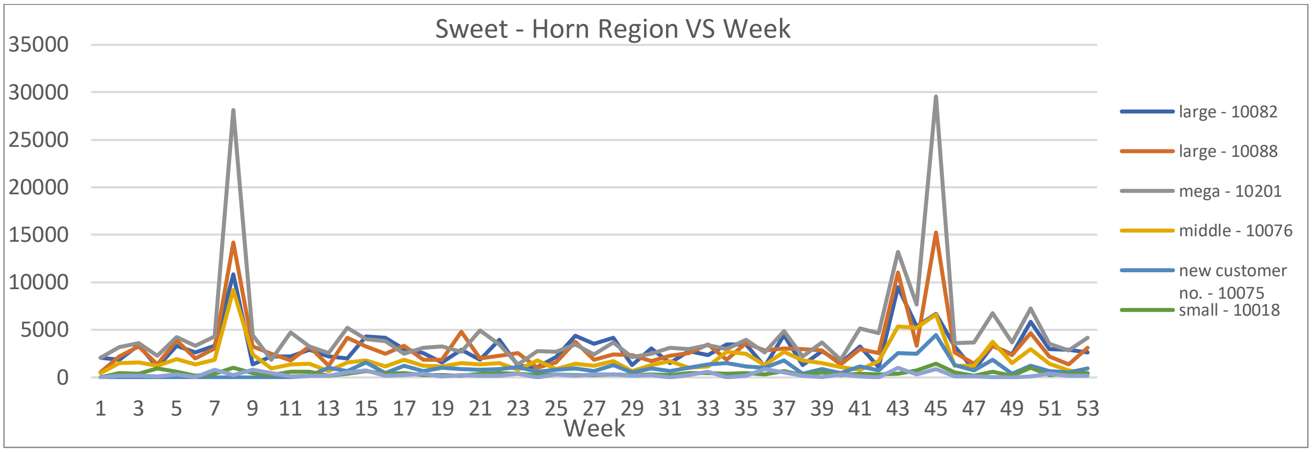

Figure 12 Sweet - Horn Region VS Week

When observing sweet turnover without VAT data across regions, no unique trends were found. However, one standout finding shows that the Horn region showcased the highest sweet turnover among all regions, reaching nearly 30,000 units in peak turnover. Like other regions, this store in Horn experienced increased sweet turnover during Week 8 and another increase from Week 42 to Week 45. These peaks align with general trends observed across various regions.

4.2.4. Geographic Analysis - Cosmetics

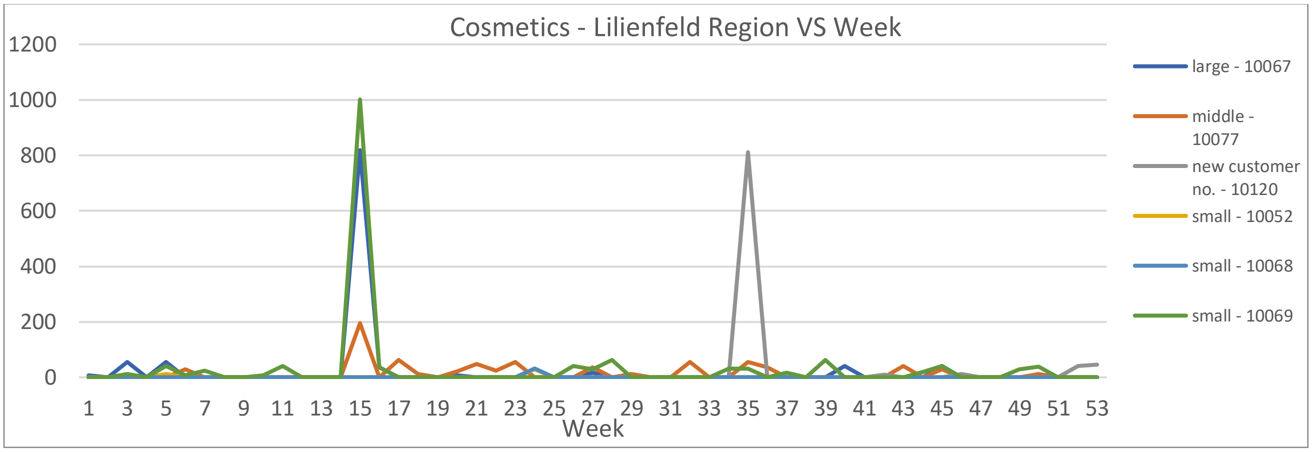

Figure 13 Cosmetics - Lilienfeld Region VS Week

In our analysis of cosmetics turnover without VAT, notable trends appear in specific weeks. In Week 15, there was a remarkable rise in cosmetics turnover at two small stores in Lilienfeld, followed by a sharp decline in the subsequent week. Similarly, during Week 35, opening a new store led to a significant spike in cosmetics turnover. However, unlike sustained growth, this increased turnover lasted only a week before dropping back to zero.

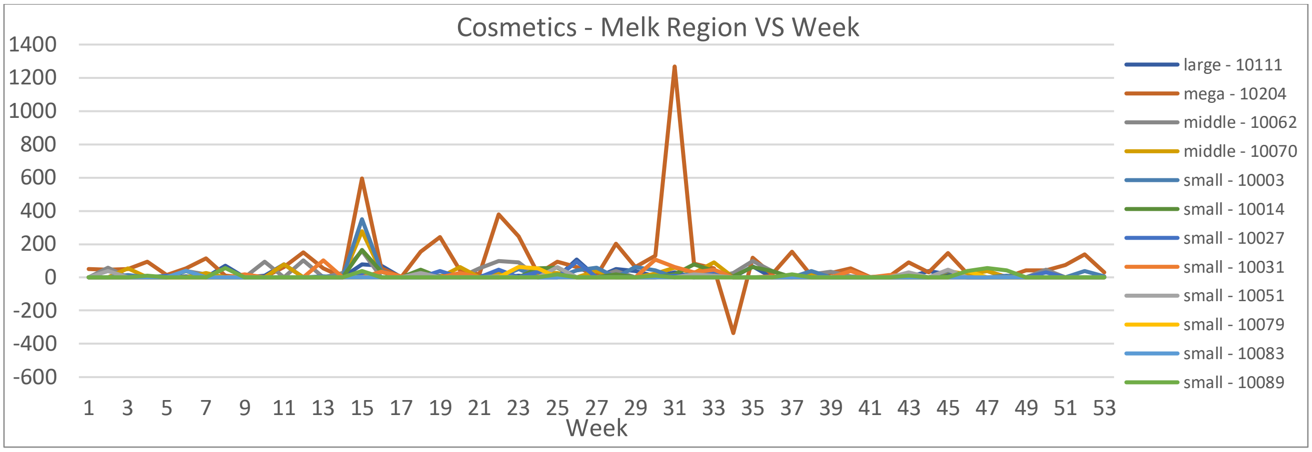

Figure 14 Cosmetics - Melk Region VS Week

In the Melk region, the cosmetics turnover without VAT data from the mega store displays a unique trend. Unlike other regions with singular peaks, mega stores in Melk experience multiple peaks in cosmetics turnover across various weeks. Among these peaks, the highest cosmetics turnover was recorded in Week 31, reaching approximately 1200 units. It has the highest turnover in cosmetics product category across all regions. However, there is an anomaly value in Week 34, where the turnover record shows a negative value. Like the Wine in Mistelbach case, we have chosen not to remove this data point immediately. Instead, we choose to confirm its accuracy before making any adjustments.

4.3. Revenue Trend Analysis

The objective of this section was to explore the annual sales data and identify growth and decline patterns. We went further to research into the reason for growth or decline patterns and this could be attributed to several factors like seasonal events that may have occurred, the opening of a new retail outlet etc. Based on our findings, we were able to make recommendations as will be discussed in the later part of this report.

4.3.1. Revenue Analysis – Wine

Based on the chart below, wine experienced consistent sales throughout the one-year period. A notable peak was seen in the month of December which can be attributed to the Christmas holiday celebration. It is expected that due to an increase in the number of events and increased shopping activity, the sales of wine (i.e., alcohol in general) will increase.

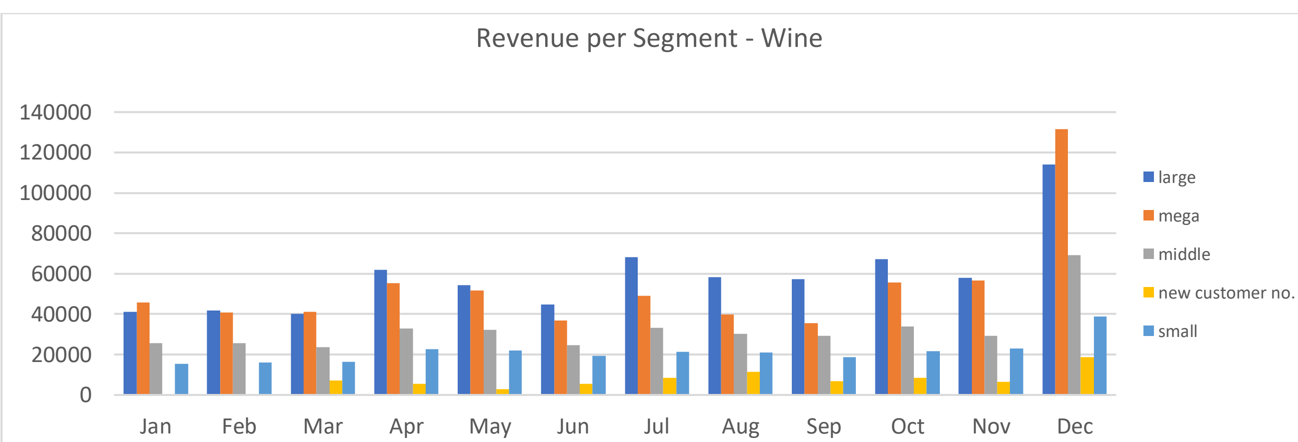

Figure 15 - Revenue per Segment - Wine

To have a clearer picture of the growth and decline patterns, we went a step further to calculate the monthly revenue growth rate using the sales data provided. Based on the graph as seen in Figure 15, we observed that all segments experienced growth in April and December and this was easily attributed to the easter and Christmas holiday celebrations. A decline was observed between May and June across all segments and while there was no major incident that occurred, we believe that the decline is expected due to the peak sales experienced in April because of the Easter holiday. Based on this decline, we also assumed that promotional offers might have been adopted to improve sales and as a result, another peak was experienced in the month of July. The peak in July was followed by a decline in August which was observed to be the trend for the year.

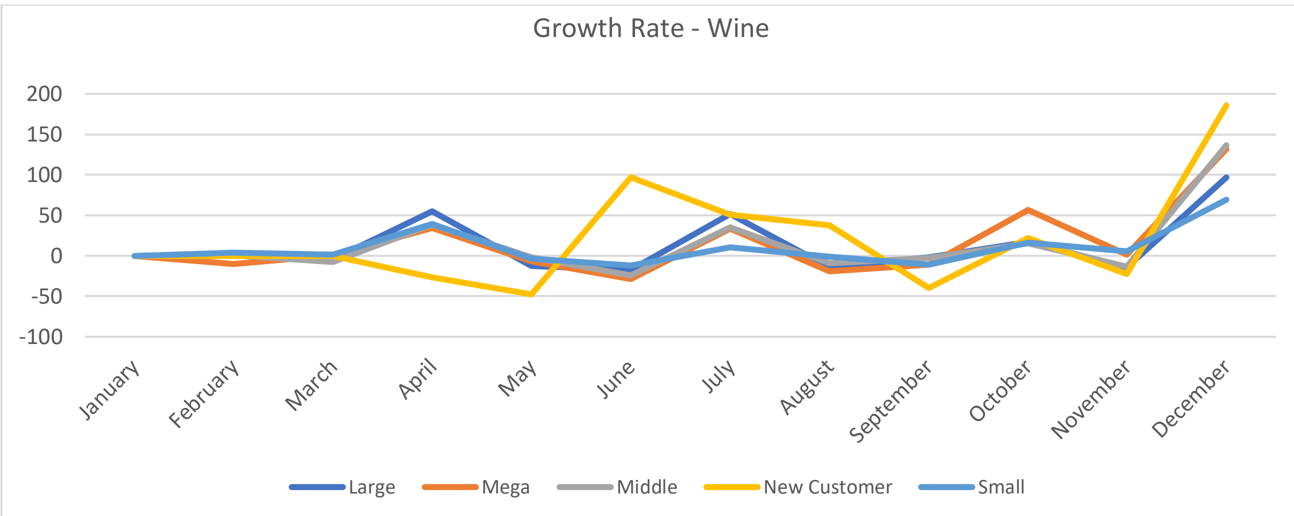

Figure 16 - Monthly Growth Rate for Wine

4.3.2. Revenue Analysis – Sweets

Amongst the three products we analysed, sweets were evidently the best seller for the one-year period. It experienced several peaks as seen in Figure 17 below. It experienced several peaks throughout the year. In February and April, the large, mega, and middle segments experienced peaks which could be attributed to Valentine’s Day and easter celebrations respectively. Also, it experienced peaks in April, October, November, and December which can be attributed to sales for holidays like Easter, Halloween, and Christmas.

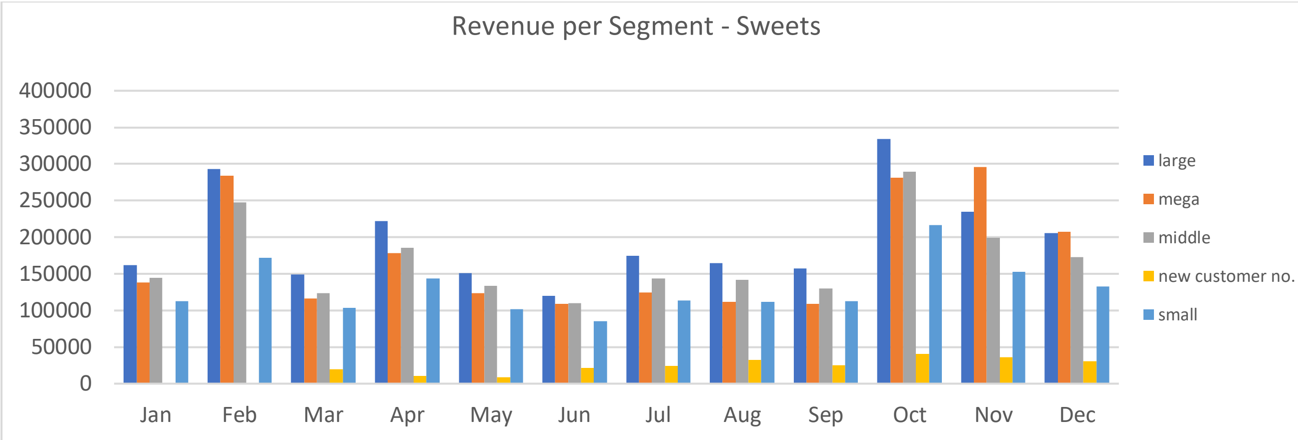

Figure 17 - Revenue per Segment for Sweets

4.3.3. Revenue Analysis - Cosmetics

Amongst all the products we chose, cosmetics was observed to be the least performing. Despite this, two major peaks occurred in April and in July. In April, the large, mega, middle, and small customers had major sales and we have a working assumption that this is due to easter holiday sales. In July, although the large, mega, middle, and small experienced peaks, the mega segments' sales for the month were way above the rest of the segments. Although there was no holiday in this month, we observed from the data that three mega customer numbers 10204, 10107 and 10202 made significant sales (1,268.20, 732.13 and 679.17 respectively) compared to the rest for the month of July. Further investigation on what happened in this specific mega segment locations is recommended.

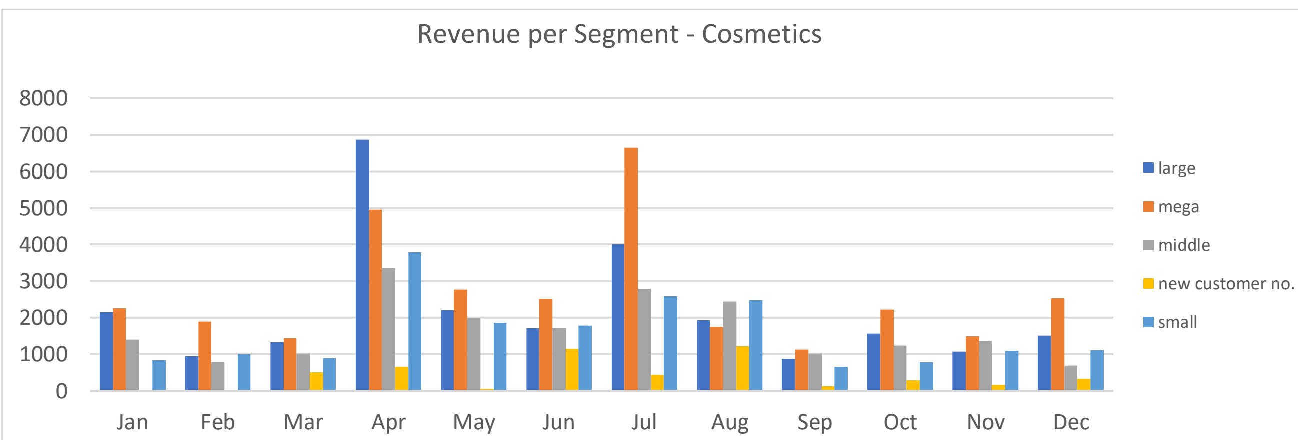

Figure 18 - Revenue per Segment for Cosmetics

Figure 19 shows the growth rate for cosmetics and confirms our observation of this product generating the least revenue in comparison to the three products being analysed.

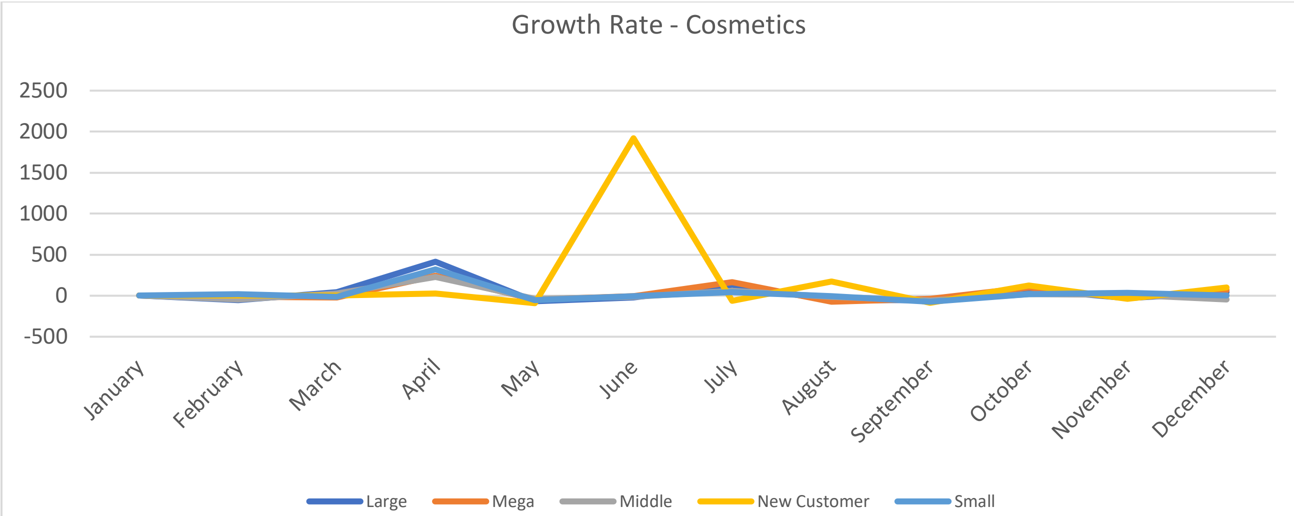

Figure 19 - Monthly Growth Rate – Cosmetics

5. RECOMMENDATIONS

In this section, we discuss the recommendations for business improvement for SPAR. The recommendations will be discussed in two parts, the recommendations based on insights from the data and general recommendations.

5.1. Optimizing Sales for future growth

In the weeks leading up to the New Year, SPAR experienced an increase in wine purchases across all customer segments, including new customers. To boost future sales of wine and sweets, SPAR could consider promoting their best products through advertisements and social media, offering discounts for special occasions. Implementing combo deals, such as discounts on the purchase of two bottles, could be another strategy. Additionally, utilizing customer preferences data to recommend wines aligning with individual tastes may enhance sales and customer satisfaction.

SPAR Austria Group observed a notable rise in cosmetics sales during specific weeks (14th to 17th week & 27th to 31st week), likely tied to holidays and seasonal shifts. To leverage this trend, the company can introduce focused promotions, both in-store and online, during these weeks. Incentives such as exclusive discounts and bundled deals may stimulate increased customer purchases. Additionally, investing in staff training to offer informed advice on seasonal trends can elevate the overall customer experience. These strategies are designed to optimize sales during peak periods and cultivate lasting customer loyalty.

5.2. Leveraging Waidhofen/t's Success and Unlocking Growth Opportunities

Analysis reveals that Waidhofen/t consistently outperforms Zwettl in sales throughout the year, indicating a stronger market presence. Spar should capitalize on Waidhofen/t's success by conducting performance analysis to understand the reason behind their success. SPAR can use the results of the analysis as a reference for their future business strategies that will be carried out in other regions. Spar should also conduct a comprehensive market study in low sales regions to identify factors influencing lower sales and tailor its product range to meet the specific demands of the local customer base.

Spar should focus on enhancing its visibility in low sales regions through community engagement and sponsorships. Building strong local ties can foster customer loyalty and drive sales growth in the long term. To improve sales in other regions, such as Zwettl, Spar should analyse pricing strategies and consider offering localized promotions or discounts that align with the preferences and purchasing behaviour of specific region residents. This will make Spar a more attractive option for consumers in the area. Additionally, Spar should invest in staff training to ensure excellent customer service in SPAR regions. Positive shopping experiences can contribute significantly to customer retention and satisfaction, ultimately benefiting sales in all regions.

5.3. Inventory Control

In response to the turnover rates, which indirectly reflect item consumption and demand within each product category and specific segment, SPAR has the capability to strategically adjust its inventory. This adaptive inventory management encompasses stock items and alternatives. The process involves leveraging historical patterns and evaluating current market scenarios, thereby enhancing the organization's forecasting capabilities.

Analysis of consumer behaviour throughout the year, considering seasonal and festival time periods.

5.4. Data Analytics and Improvement Strategies

From the data analysis, SPAR identified negative data input within several regions, such as the Melk region in week 34. This is an issue that needs immediate attention. To address this issue, here are some recommendations that SPAR can implement to optimize operations:

- SPAR can conduct a comprehensive root cause analysis to identify the factors why the performance is negative in various regions, such as Melk during week 34. It is recommended to investigate the reason behind the issue, whether anomalies are due to a technical glitch, data entry errors, or external factors.
- SPAR should establish a real-time monitoring system that will detect anomaly data and deviations from expected performance. For taking immediate action SPAR can implement automated alerts that will notify stakeholders when unusual patterns or deviations from expected data occur.

5.5. Strategies for year-round growth in Cosmetics sales

SPAR has experienced a consistent challenge and low sales for cosmetics throughout the year except during the Easter season in April, and the summer vacation period in July. They can conduct a cost analysis within the cosmetic sector. SPAR should identify areas for potential optimization by researching the current cost structure without compromising the quality of products. They can negotiate favourable terms with cosmetic suppliers and the exploration of innovative yet cost-effective marketing strategies. Making a balance between cost reductions and maintaining product quality can give a boost to the profitability in the cosmetics segment for SPAR.

5.6.

5.7. Recommendations (General)

5.7.1. Customer Feedback Collection for Continuous Improvement

To enhance customer satisfaction and identify areas for improvement, SPAR should collect feedback from customers. Implementing feedback mechanisms like surveys, online reviews, and in-store suggestion boxes. SPAR will gain valuable insights into customer preferences and expectations. The collected data should be analysed regularly to figure out patterns and identify areas of improvement. By incorporating customer feedback, SPAR can cultivate a customer-centric approach that will help SPAR maintain the highest customer satisfaction.

5.7.2. Exclusive Product Collaborations

SPAR can explore collaborations with local and renowned brands to create exclusive products that include sweets, wines, cosmetics or other product groups, providing provide an exclusive and distinctive offer that will not be available anywhere. For example, SPAR can collaborate with a local chocolatier to create a limited-edition line of chocolate that pairs perfectly with a selection of wines stocked exclusively at SPAR. This exclusive collaboration could also extend to a unique line of complementary cosmetics, such as scented candles or skincare products.

5.7.3. Digital Engagement

SPAR can increase their interaction using digital platforms. They can build a separate small team to create engaging content and promotions showing sweets, wines, cosmetics, or other product groups complementing each other. Taking advantage of modern social platforms and online ads and promotions channels can make sure SPAR's unique products are in the spotlight. By doing this, SPAR can boost its visibility, attract more customers, and drive both online and in-store sales.

5.7.4. Personalized Customer Recommendations

SPAR can leverage customer purchase history and preference to provide personal recommendations, enhancing the shopping experience. This will also help determine which products should be included more in the cross-category purchase section to attract more customers and increase sales.

6. CRITICAL REFLECTION & LIMITATIONS

This chapter discusses the assumptions made during the analysis, the limitations of the process, and any flaws or problems that occur during the development of the work.

6.1. Assumptions

During the analysis, there are several assumptions that was made. These assumptions is the anomaly turnover value, No special interest, relationship between the data and the population data.

6.1.1. Anomaly Turnover Value

Negative values were found that we saw that could affect the results. We debated whether to remove these values or not and finally decided not to edit any of them. Because of this, assumptions were made. We noticed that in most of the cases where this occurred just before there had been a high or prominent spike, so the assumption was made that it could be product returns or something similar. In the case of wine or sweets it does not make much sense to make returns, but in the case of cosmetic products it does, and that is where these negative values have been found. Not only negative value, there are high values that also suspicious because there are some products with the high turnover for one week and it does not make another increase in the other weeks. We already tried to find the answer however there are not any possible explanation about it. As a result, we decided to not change this value and assume this value is correct.

6.1.2. No Special Interest

Another of the assumptions made is that no special interest has been paid to important times of the year such as Christmas, when it is taken for granted that sales of most products will increase. Therefore, an attempt has been made to analyse in greater depth situations that might attract attention, such as peaks on non-remarkable dates or peaks that do not follow the previous patterns.

6.1.3. Correlation between Data

It was often difficult to find a relationship with the data to be reviewed and to justify why some peaks often appeared. That is to say, finding correlation with some data without knowing for example if there was some kind of offer or promotion on those dates was a bit difficult. And looking for that kind of information to justify some sales was even more difficult.

6.1.4. Population Data

The given data is the turnover of SPAR supermarkets in 2009. However, the data population that our group used is the population of the lower Austria in 2023. Our group decided to use this data because we cannot find the complete population data in lower Austria in 2009. As a result, we assumed that the population ratio of the lower Austria is same between 2009 and 2023.

6.2. Limitations

In the process of work development, our group found several limitations that reduce the analysis result that can be presented. This limitation occurs because our group does not have access to the data centre of the SPAR supermarket and unable to confirm about some anomaly in the given data. Here are some limitations that our group found:

6.2.1. Obtaining Additional Data

The given data is only providing several basic information about SPAR supermarket turnover in lower Austria. As a result, it limits the output that are able to provide by our group and also limits the variety of data that can our group analyse. For example, in the given data, there are not any values that relate to the net profit. If we have this kind of data, there is a possibility that our group will able to provide another analysis that might be able to help the company. The other example, there are not any previous year data. If this kind of data is provided, it would be easier for our group to make a forecast analysis and provided the business strategic for future.

6.2.2. Data Validity

During the analysis process, our group was made several assumptions that relate to the data. These assumptions are made due to our group was unable to confirm the accuracy of the data. One of the biggest assumptions that our group made are the negative values in the data. These assumptions can lead to a misinformation and makes the reader or stakeholder takes the wrong business strategic.

6.3. Problem

This subsection discusses about the problem that occurs during the process of analysis. There are three major problem that happen during our development of the work, data validity, competencies, and lack of data variation.

6.3.1. Data Validity

The data validity created a biggest problem in our process of working with the SPAR supermarket data. It is because we found several anomaly data like quite big negative turnover value. This kind of value probably would not be happened in the practice. Our group are also unable to confirm the data accuracy to SPAR supermarket. This led us to assume that might be happened. After a long discussion with our team, we decided to not change the negative value and report it in our presentation. We choose this method because it is safer than we make any change and created a misinformation in our analysis.

6.3.2. Competencies

The second major problem that occur in our group is competencies. Our group consist of people from various fields but there are no anyone who have a professional experience in retail. This makes our group making some assumptions to create a desirable output, like we assume that the stakeholder want to know the trend that happened in certain product group during the year or which regions that has a unique turnover in 2009.

6.3.3. Lack of Data Variation

As discussed in section 6.1 and 6.2, the lack of data variation is created a problem when our group want to analyse the SPAR supermarket further. For example, we cannot create a forecast analysis because we do not have any data of previous year, or we cannot make ROE. However, we tried to cover the problem by searching for additional data that we can found in the internet such as population data and Austria traditional events.

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